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1 Common (and less common) Graphs for EDA

One of the major advantages of the grammar of graphics — especially as implemented in the tidyverse package `ggplot2` — is that it enables us to rapidly create *and re-create* graphs. This speed and flexibility is extremely useful for EDA.

```
# install.packages(c('tidyverse', 'skimr', 'gghighlight', 'ggbeeswarm', 'ggforce', 'ggupset'))
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.0 --

## v ggplot2 3.3.2     v purrr   0.3.4
## v tibble  3.0.3     v dplyr   1.0.2
## v tidyr   1.1.2     v stringr 1.4.0
## v readr   1.3.1     v forcats 0.5.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```
library(skimr)

library(ggbeeswarm)
library(ggforce)
# library(gghighlight)
# library(ggupset)
```

While `ggplot2` is supposed to have "pretty defaults," for a long time its default color scales have been *terrible*, with bad color combinations for color-blind folks, poor rendering in black and white, and low contrast against the default gray background.

The options below should set a more color-blind-friendly default palette and a plain white background.

```
options(ggplot2.discrete.fill = partial(colorspace::scale_fill_discrete_qualitative, palette = 'Dark 2'
        ggplot2.discrete.colour = partial(colorspace::scale_color_discrete_qualitative, palette = 'Dark

theme_set(theme_bw())
```

1.1 Data

We'll be looking at a collection of US college tuition and diversity data. I acquired the data from TidyTuesday (2020-03-10), which in turn sourced it from some combination of Tuition Tracker and the *Chronicle of Higher Education*.

The documentation in this TidyTuesday repo is quite poor, and doesn't actually say exactly where each dataset came from. Some of the data is parsed from downloaded HTML files; other data is produced by non-reproducible scripts that clean CSV files. My best guess is that the tuition data comes a *Chronicle of Higher Education* table with AY 2018-19 "sticker prices" for tuition. And I think the diversity data comes from another *CHE* table which reports counts of all students (not just undergrads) enrolled in Fall 2014.

To facilitate some later steps, I filtered the datasets we'll be using to *4-year colleges and universities* in *California*. I also standardized the names of the Cal State (California State University, or CSU) and University of California campuses, and identified which "system" each school is in (CSU, UC, private, or for-profit). The prepared data files (along with the cleaning script and draft versions of the plots below) is available here.

```
data_dir = 'data'
data_files = file.path(data_dir, c('diversity.csv', 'tuition.csv'))

data_urls = str_c('https://github.com/data-science-methods/ca-tuition-diversity/raw/master/', data_files

## `if` isn't vectorized, so we have to loop over the files/URLs manually
download_file = function(path, url) {
    if (!file.exists(path)) {
        download.file(url, path)
    }
}
purrr::walk2(data_files, data_urls, download_file)

div_df = read_csv(data_files[1])
```

```
## Parsed with column specification:
## cols(
##   name = col_character(),
##   total_enrollment = col_double(),
##   state = col_character(),
##   category = col_character(),
##   enrollment = col_double(),
##   share = col_double()
## )
```

```
tuition_df = read_csv(data_files[2])
```

```
## Parsed with column specification:
## cols(
##   name = col_character(),
##   state = col_character(),
##   state_code = col_character(),
##   type = col_character(),
##   degree_length = col_character(),
##   room_and_board = col_double(),
##   in_state_tuition = col_double(),
##   in_state_total = col_double(),
##   out_of_state_tuition = col_double(),
##   out_of_state_total = col_double(),
##   system = col_character()
## )
```

`tuition_df` contains data on tuition (the "sticker price," not what students actually pay). There are 135 rows, 1 row for each of the public, private, and for-profit 4-year schools in California. There are a few different calculations of tuition or cost; we'll only be looking at `in_state_tuition`. In this dataset, `type` is either

public, private, or for-profit. All of the public 4-year colleges or universities in California belong to either the Cal State or UC system. `system` distinguishes these two public systems, and also carries over the values from `type` for private and for-profit schools.

```r
skim(tuition_df)
```

```
## -- Data Summary ------------------------
##                           Values
## Name                      tuition_df
## Number of rows            135
## Number of columns         11
## _______________________
## Column type frequency:
##   character               6
##   numeric                 5
## ________________________
## Group variables           None
##
## -- Variable type: character -----------------------------------------------------
##   skim_variable n_missing complete_rate   min   max empty n_unique whitespace
## 1 name                  0             1    13    59     0      135          0
## 2 state                 0             1    10    10     0        1          0
## 3 state_code            0             1     2     2     0        1          0
## 4 type                  0             1     6    10     0        3          0
## 5 degree_length         0             1     6     6     0        1          0
## 6 system                0             1     2    10     0        4          0
##
## -- Variable type: numeric -------------------------------------------------------
##   skim_variable        n_missing complete_rate   mean     sd    p0    p25    p50
## 1 room_and_board              35         0.741 12865.  3288.  2800 10932. 13428.
## 2 in_state_tuition             0         1     25602. 16268.  2691 10882. 20670
## 3 in_state_total               0         1     35132. 19484.  2691 19377  30416
## 4 out_of_state_tuition         0         1     29425. 14629.  2691 17531  27810
## 5 out_of_state_total           0         1     38955. 19215.  2691 23102  36103
##      p75  p100 hist
## 1 15332. 18127
## 2 40172. 56876
## 3 51009  75003
## 4 43113  56876
## 5 57560  75003
```

```r
count(tuition_df, type, system)
```

```
## # A tibble: 4 x 3
##   type       system         n
##   <chr>      <chr>      <int>
## 1 For Profit for profit    10
## 2 Private    private       93
## 3 Public     CSU           23
## 4 Public     UC             9
```

`div_df` has 5027 rows, corresponding to unique combinations of school (`name`) and demographic group (`category`). `share` gives the fraction of students enrolled at the school in the given group. Because the TidyTuesday dataset didn't have the `degree_length` variable, I wasn't able to filter down to only 4-year schools. That won't be a problem later.

```r
skim(div_df)
```

```
## -- Data Summary ------------------------
##                           Values
## Name                      div_df
## Number of rows            5027
## Number of columns         6
## _______________________
## Column type frequency:
##   character               3
##   numeric                 3
## ________________________
## Group variables           None
##
## -- Variable type: character -----------------------------------------------------
##   skim_variable n_missing complete_rate   min   max empty n_unique whitespace
## 1 name                  0             1    11    74     0      457          0
## 2 state                 0             1    10    10     0        1          0
## 3 category              0             1     5    34     0       11          0
##
## -- Variable type: numeric -------------------------------------------------------
##   skim_variable    n_missing complete_rate     mean       sd    p0      p25
## 1 total_enrollment         0             1 5912.    8886.        8 427
## 2 enrollment               0             1 1160.    2988.        0  14
## 3 share                    0             1    0.193    0.241     0   0.0111
##         p50      p75  p100 hist
## 1 1232      8695     51237
## 2   98       580     36116
## 3    0.0634    0.332     1
```

```r
count(div_df, name)
```

```
## # A tibble: 457 x 2
##    name                                              n
##    <chr>                                         <int>
##  1 Academy for Jewish Religion                      11
##  2 Academy of Art University                        11
##  3 Academy of Chinese Culture and Health Sciences   11
##  4 Academy of Couture Art                           11
##  5 Acupuncture and Integrative Medicine College     11
##  6 Advanced College                                 11
##  7 Advanced Computing Institute                     11
##  8 Advanced Training Associates                     11
##  9 Allan Hancock College                            11
## 10 Alliant International University: San Diego      11
## # ... with 447 more rows
```

```r
count(div_df, category)
```

```
## # A tibble: 11 x 2
##    category                              n
##    <chr>                             <int>
##  1 American Indian / Alaska Native     457
##  2 Asian                               457
##  3 Black                               457
```

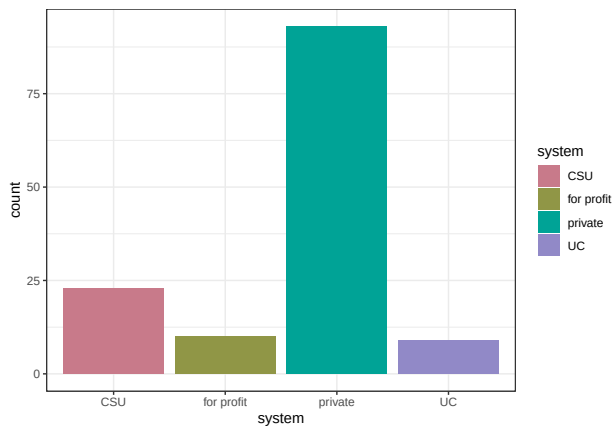
```
##  4 Hispanic                              457
##  5 Native Hawaiian / Pacific Islander    457
##  6 Non-Resident Foreign                  457
##  7 Total Minority                        457
##  8 Two Or More Races                     457
##  9 Unknown                               457
## 10 White                                 457
## 11 Women                                 457
```

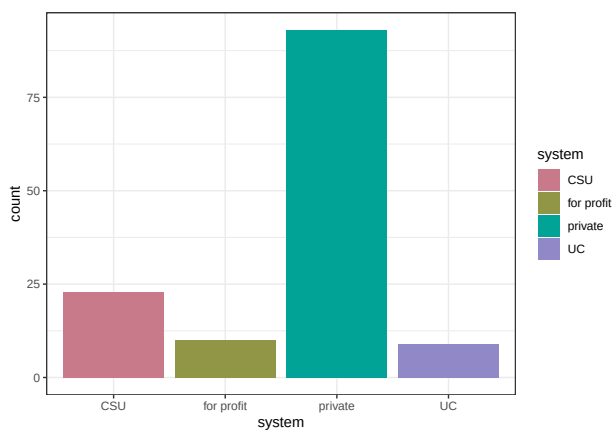
1.2 1D distributions

1.2.1 Categorical: Barplot

Counts of observations for a single categorical variable. It's a bar geom with a count statistic. If we keep the statistic but change the geom, we get a point version of a barplot.

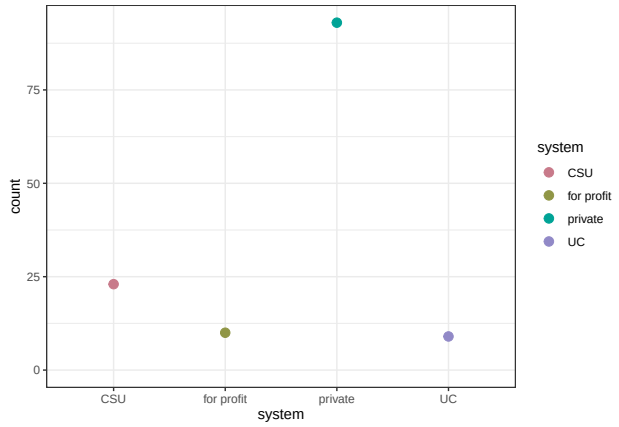
```
ggplot(tuition_df, aes(system, fill = system)) +
    geom_bar()
ggplot(tuition_df, aes(system, fill = system)) +
    stat_count(geom = 'bar')
```

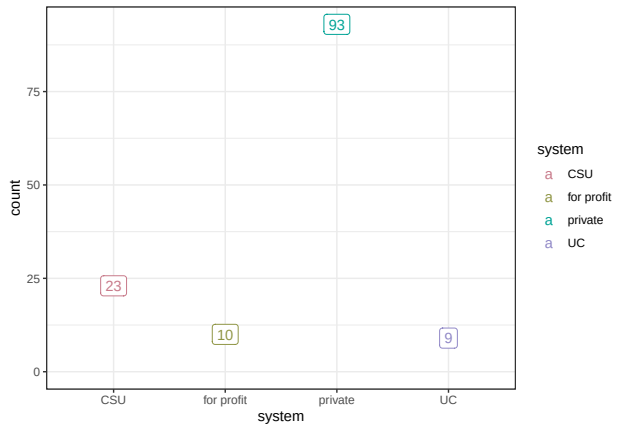




If we keep the statistic but change the geom, we can make a point version of a barplot. Or a version with text labels.

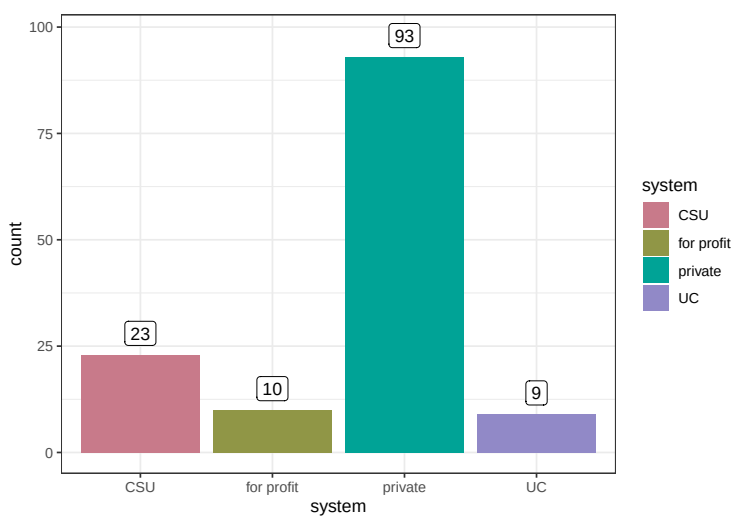
```
ggplot(tuition_df, aes(system, color = system)) +
    stat_count(geom = 'point', size = 3)
ggplot(tuition_df, aes(system, color = system)) +
    stat_count(geom = 'label', aes(label = after_stat(count)))
```





Because of ggplot's layering system, we're not restricted to just a barplot or one of these variations. We can combine geoms into a single plot.

```
ggplot(tuition_df, aes(system)) +
    stat_count(geom = 'bar', aes(fill = system)) +
    stat_count(geom = 'label', aes(label = after_stat(count)),
               position = position_nudge(y = 5))
```



1.2.2 Numeric: Histograms and density plots

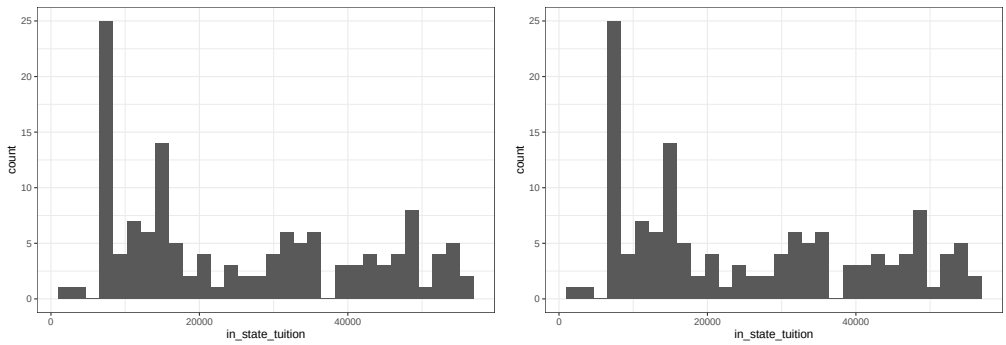
A histogram is a count of observations for a single categorical variable. It uses the same geom as the barplot, but a statistic of counts of *binned* values for a continuous variable.

```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

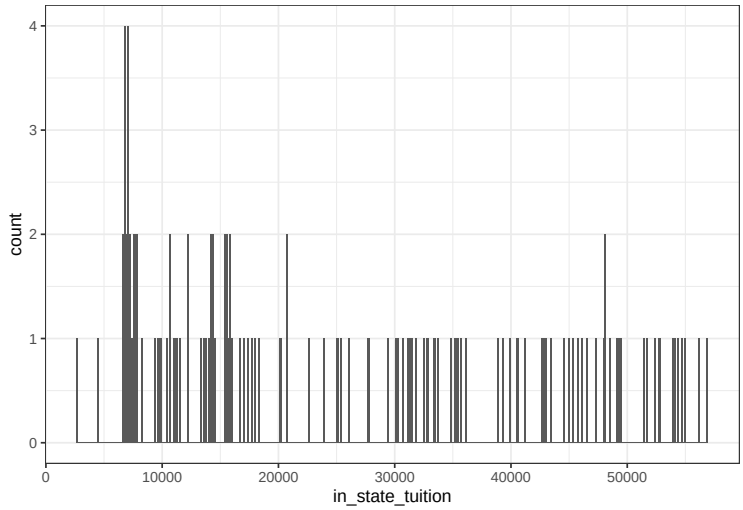
```
ggplot(tuition_df, aes(in_state_tuition)) +
    stat_bin(geom = 'bar')
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

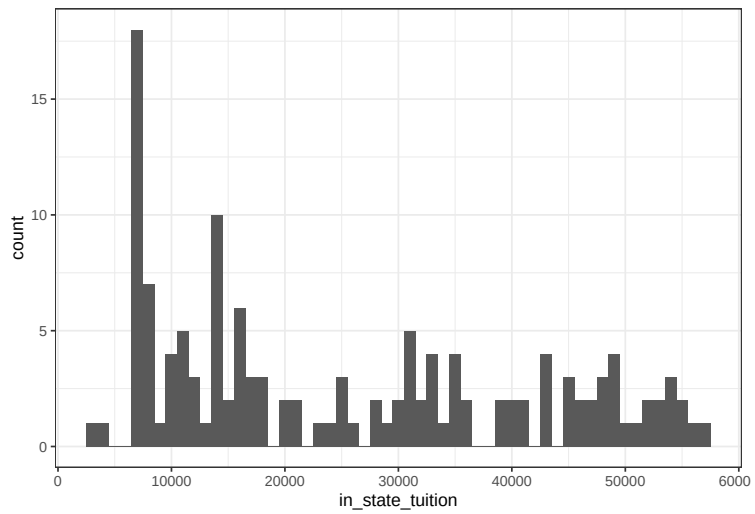


- `binwidth` can have a dramatic effect on the appearance of the plot
- `binwidth` is a free parameter in constructing the histogram as a model of the data

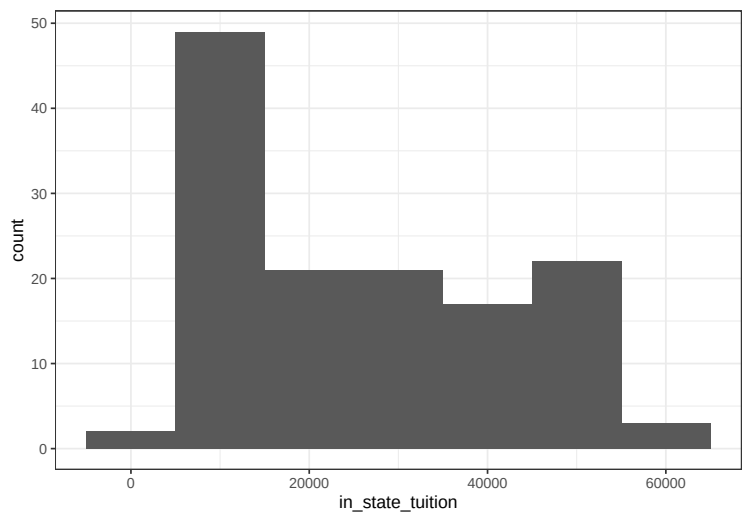
```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_histogram(binwidth = 100)
```




```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_histogram(binwidth = 1000)
```



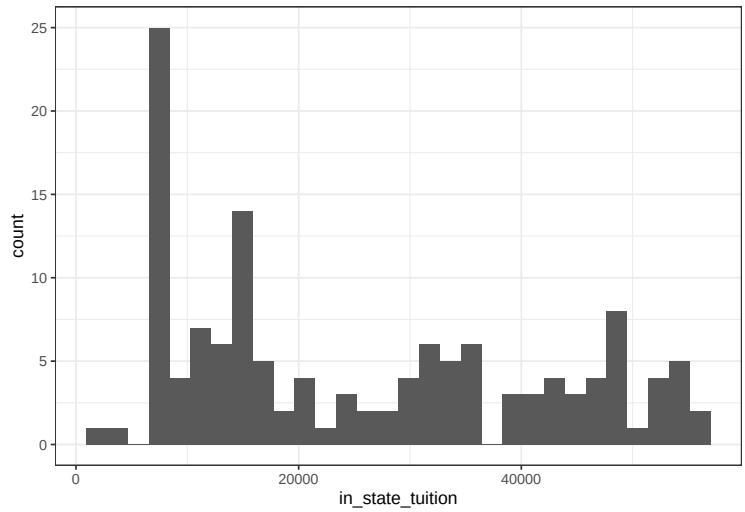

```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_histogram(binwidth = 10000)
```



A frequency polygon uses the same count statistic, but displays the result with a line rather than columns.

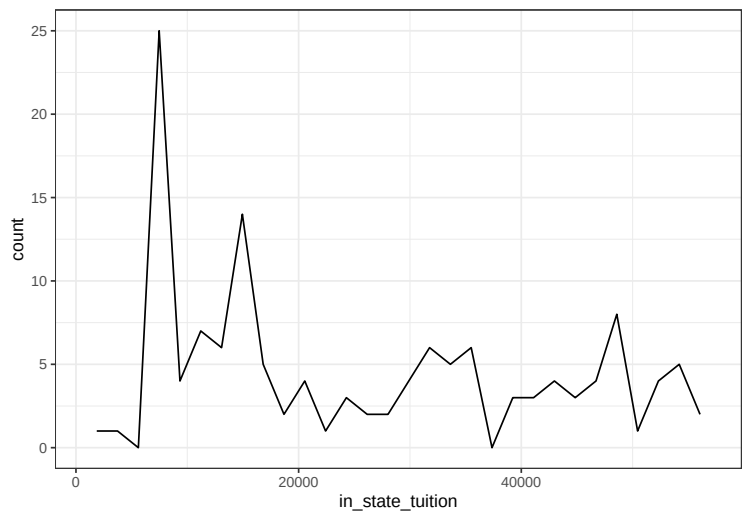
```
ggplot(tuition_df, aes(in_state_tuition)) +
    stat_bin(geom = 'bar')
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



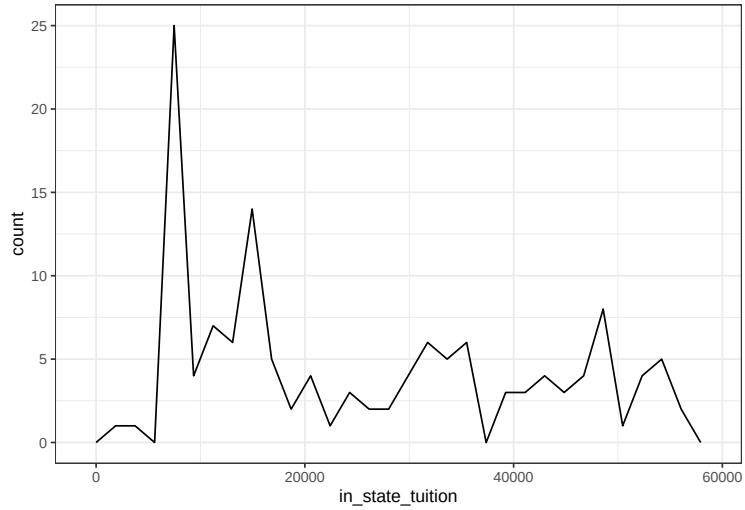

```
ggplot(tuition_df, aes(in_state_tuition)) +
    stat_bin(geom = 'line')
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_freqpoly()
```

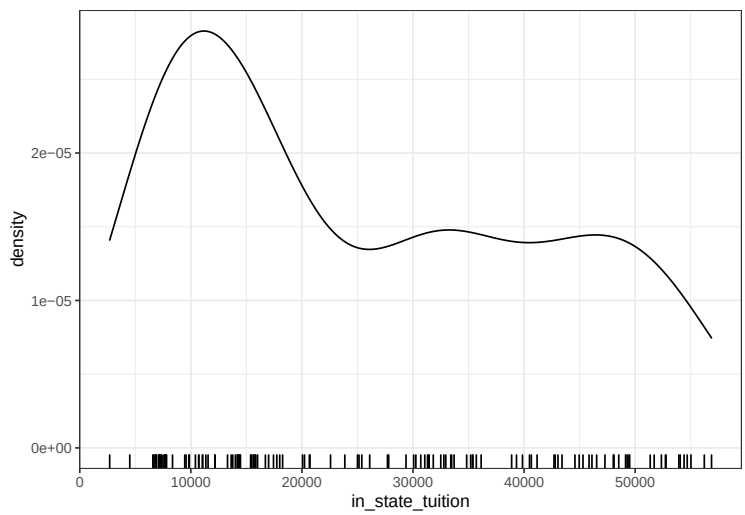
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Density plots are a smoothed version of histograms.

- Often drawn with a rug, which indicates the locations of observed values on the x-axis
- Like histogram, is a model of the data
- Unlike histogram, people often overlook the free parameters (see `?density`)

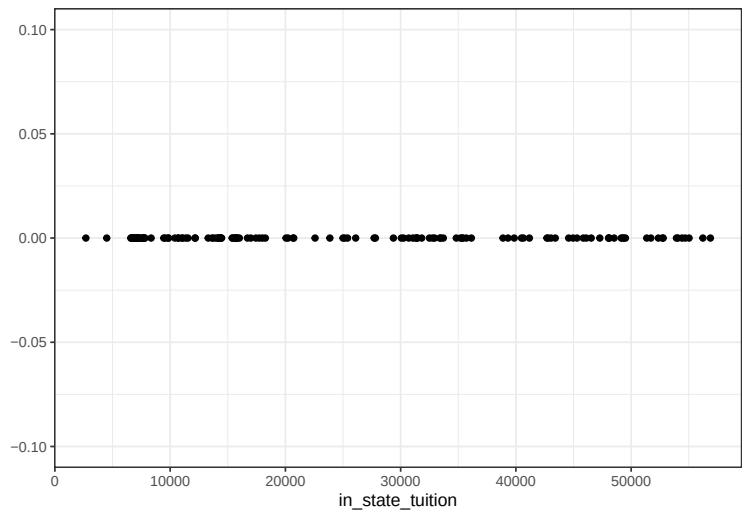
```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_density() +
    geom_rug()
```



A strip plot simply plots the values on a single axis.

- Unlike all of the other examples here, it doesn't involve any free parameters or other modeling decisions.
- But it's also not very useful

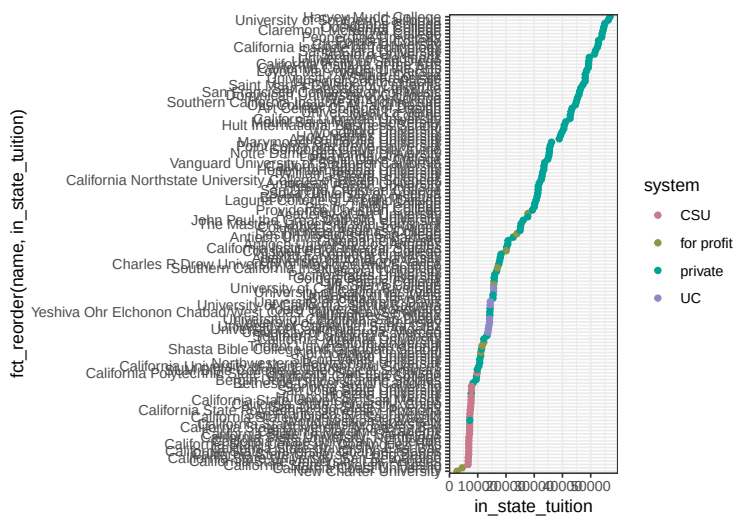
```
ggplot(tuition_df, aes(in_state_tuition)) +
    geom_point(y = 0) +
    ylim(-.1, .1)
```



1.3 2D distributions: Categorical x Numeric

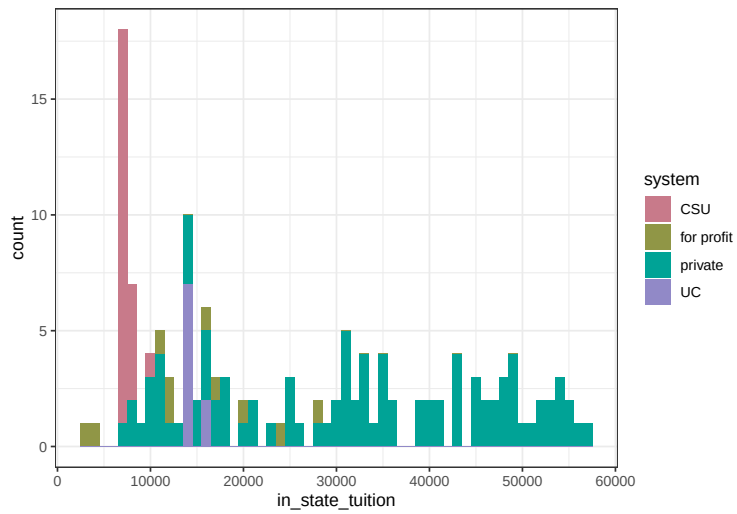
When there are only a few values of a categorical variable, it can be useful to simply plot points for individual values.

```
ggplot(tuition_df, aes(fct_reorder(name, in_state_tuition),
                       in_state_tuition,
                       color = system)) +
    geom_point(aes(color = system)) +
    coord_flip()
```



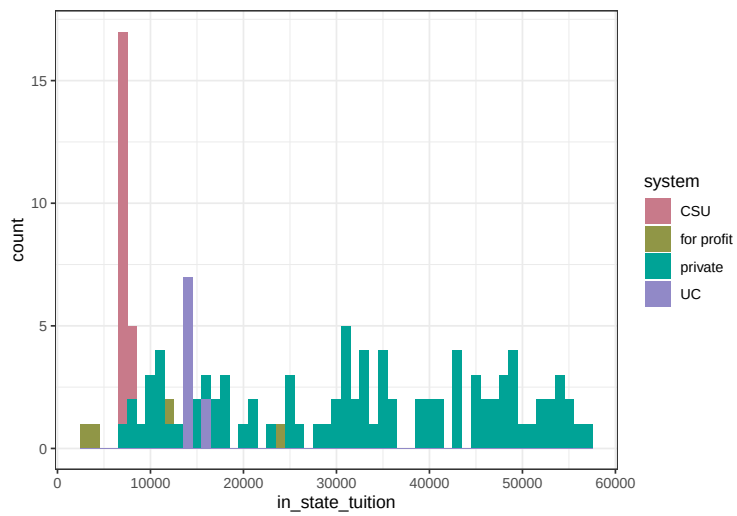
But we probably don't care about the names of individual schools. One option is a stacked histogram, using color (fill) to indicate school system.

```
ggplot(tuition_df, aes(in_state_tuition, fill = system)) +
    geom_histogram(binwidth = 1000)
```



These can be difficult to read when it's not clear whether the bars are stacked on top of each other or layered, one behind the other. Here the bars are layered. The differences are very subtle but potentially important.

```
ggplot(tuition_df, aes(in_state_tuition, fill = system)) +
    geom_histogram(binwidth = 1000, position = 'identity')
```



So it's more common to put the categorical variable on the x-axis. Here's a traditional boxplot. While color is strictly unnecessary, using color for the same categorical variable across plots helps us track relationships. Notice that we can change the way color is used by simply changing which aesthetic gets `system`.

```
ggplot(tuition_df, aes(system, in_state_tuition, color = system)) +
    geom_boxplot()
```

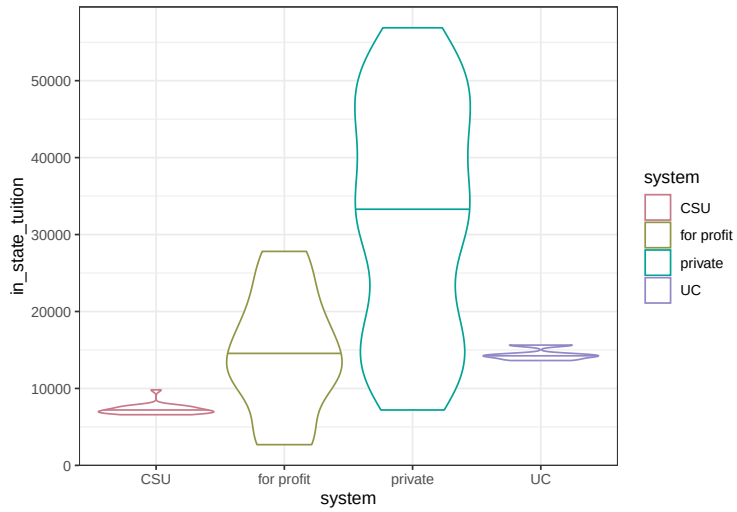
```
ggplot(tuition_df, aes(system, in_state_tuition, fill = system)) +
    geom_boxplot()
```

However, bowplots have many of the problems that @WeissgerberBarLineGraphs2015 identify with bar-and-whisker plots. Neither is actually very good at showing non-gaussian distributions of the underlying data.

A violin plot is better for this. The violin plot calculates densities across the numerical variable (tuition) within groups defined by the categorical variable (system), and shows them symmetrically. We can use the

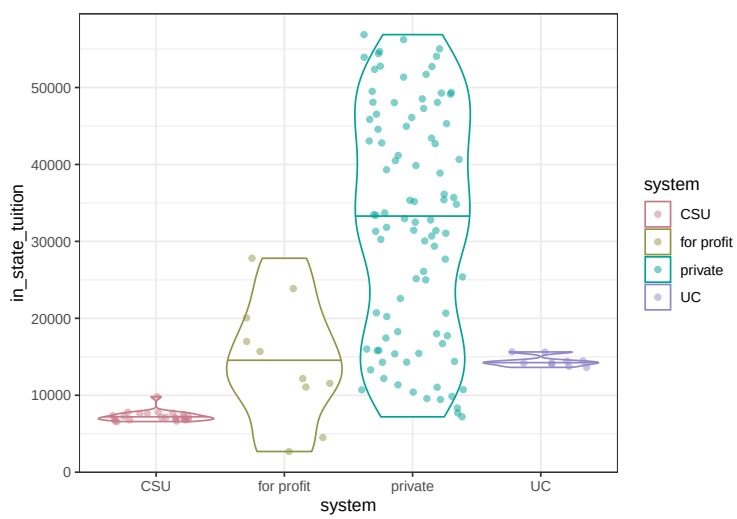
`draw_quantiles` argument to show the medians, and `scale = 'width'` to get them to have the same width (rather than the same area, the default).

```
ggplot(tuition_df, aes(system, in_state_tuition, color = system)) +
    geom_violin(draw_quantiles = .5, scale = 'width')
```



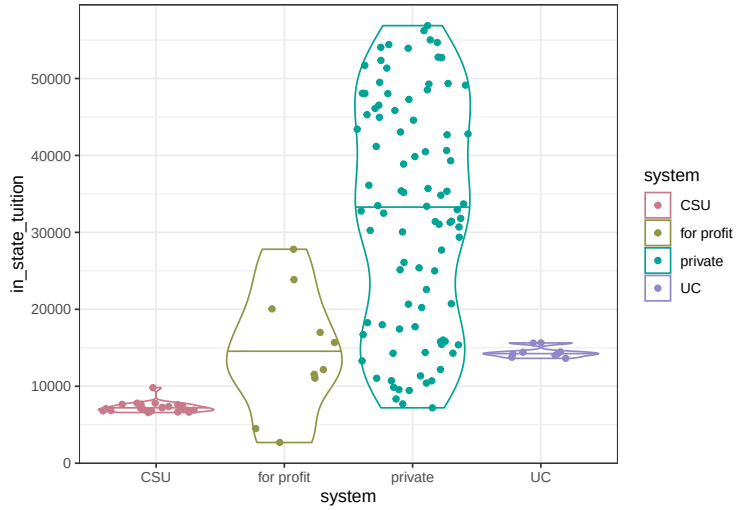
We can add points to this, using `position = 'jitter'` to reduce overplotting and `alpha = .5` to make them 50% transparent.

```
ggplot(tuition_df, aes(system, in_state_tuition, color = system)) +
    geom_violin(draw_quantiles = .5, scale = 'width') +
    geom_point(position = 'jitter', alpha = .5)
```



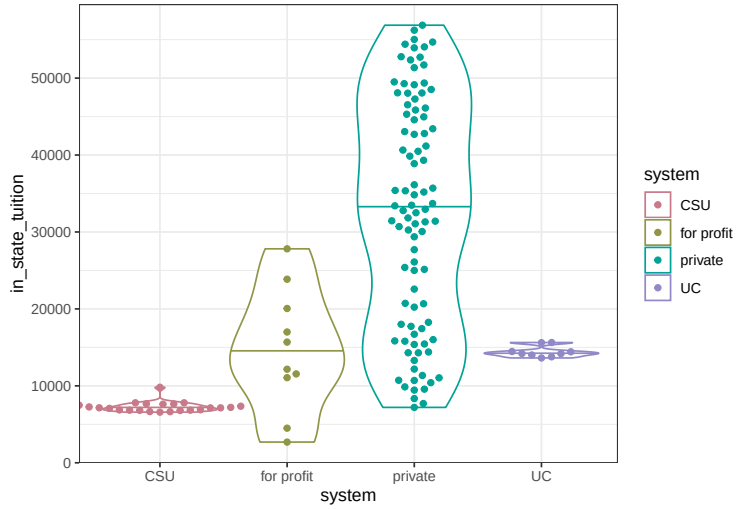
`ggforce::geom_sina()` tries to match the amount of jitter to the width of the violin plot, using the same `scale` argument. It's a little better, because now the points aren't outside of the violin. But it's still hard to see a visual pattern.

```
ggplot(tuition_df, aes(system, in_state_tuition, color = system)) +
    geom_violin(draw_quantiles = .5, scale = 'width') +
    geom_sina(scale = 'width')
```



A beeswarm plot positions the points so that they cluster close together in the x-direction without too much overplotting. This allows us to show both the individual points and the density distribution, with roughly the same shape (after tweaking some parameters).

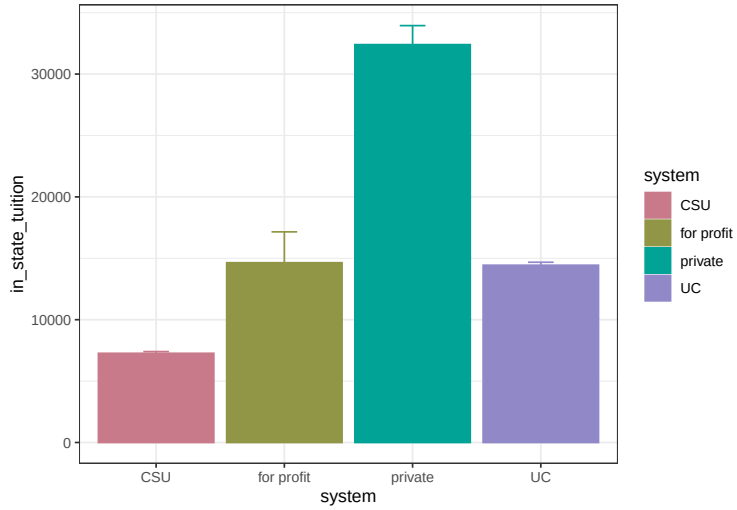
```
ggplot(tuition_df, aes(system, in_state_tuition, color = system)) +
    geom_violin(draw_quantiles = .5, scale = 'width') +
    geom_beeswarm(cex = 2)
```



Using ggplot it is possible to create the bar-and-whiskers plot that @WeissgerberBarLineGraphs2015 criticize. But now we have much better options.

```
ggplot(tuition_df, aes(system, in_state_tuition,
                       fill = system, color = system)) +
    stat_summary(geom = 'bar') +
    stat_summary(geom = 'errorbar', width = .2)
```

```
## No summary function supplied, defaulting to `mean_se()`
## No summary function supplied, defaulting to `mean_se()`
```



1.4 Joining tuition and diversity data

The other numeric variables in the tuition data are variations on in-state tuition: they add room and board (`in_state_total`) or tuition for out-of-state tuitions (`out_of_state_tuition`), or both. For something more interesting, we'll join the tuition data to the diversity data.

Recall that `div_df` has one row for each *combination* of school and demographic group (`category`).

```
div_df
```

```
## # A tibble: 5,027 x 6
##    name          total_enrollment state   category             enrollment   share
##    <chr>                    <dbl> <chr>   <chr>                     <dbl>   <dbl>
##  1 Ashford Uni~            51237 Califo~ Women                    36116 0.705
##  2 Ashford Uni~            51237 Califo~ American Indian / A~       482 0.00941
##  3 Ashford Uni~            51237 Califo~ Asian                      645 0.0126
##  4 Ashford Uni~            51237 Califo~ Black                    18727 0.365
##  5 Ashford Uni~            51237 Califo~ Hispanic                  4792 0.0935
##  6 Ashford Uni~            51237 Califo~ Native Hawaiian / P~       405 0.00790
##  7 Ashford Uni~            51237 Califo~ White                    23596 0.461
##  8 Ashford Uni~            51237 Califo~ Two Or More Races         1539 0.0300
##  9 Ashford Uni~            51237 Califo~ Unknown                    955 0.0186
## 10 Ashford Uni~            51237 Califo~ Non-Resident Foreign        96 0.00187
## # ... with 5,017 more rows
```

First we'll do some checks for the hazards discussed in the "Joins" chapter.

```
## Schools in tuition_df that will be missing after we join w/ div_df
anti_join(tuition_df, div_df, by = 'name') %>%
    pull(name)
```

```
##  [1] "Alliant International University"
##  [2] "Antioch University Los Angeles"
##  [3] "Antioch University Santa Barbara"
##  [4] "Beverly Hills Design Institute"
##  [5] "California Coast University"
##  [6] "California Northstate University College of Health Sciences"
##  [7] "Charles R Drew University of Medicine and Science"
##  [8] "Concordia University Irvine"
```

```
##  [9] "Hult International Business School"
## [10] "Laguna College of Art and Design"
## [11] "Lincoln University"
## [12] "Mount Saint Mary's University"
## [13] "New Charter University"
## [14] "Northcentral University"
## [15] "Northwestern Polytechnic University"
## [16] "Pacific Union College"
## [17] "Silicon Valley University"
## [18] "Simpson University"
## [19] "SUM Bible College & Theological Seminary"
## [20] "The Master's University and Seminary"
```

```r
## Because div_df isn't filtered down to 2-year schools, it's okay if a lot of its schools are left out
anti_join(div_df, tuition_df, by = 'name') %>%
    nrow()
```

```
## [1] 3762
```

```r
## Checking duplications
comb_df_draft = inner_join(tuition_df, div_df, by = 'name')
nrow(comb_df_draft)
```

```
## [1] 1265
```

```r
nrow(distinct(comb_df_draft))
```

```
## [1] 1265
```

```r
## 135 schools in tuition_df, minus 20 rows not matched
comb_df_draft %>%
    select(name) %>%
    distinct() %>%
    nrow()
```

```
## [1] 115
```

```r
## We should have one row in comb_df_draft for each school-category combination
comb_df_draft %>%
    select(name, category) %>%
    distinct() %>%
    nrow()
```

```
## [1] 1265
```

```r
## Check column names
names(comb_df_draft)
```

```
##  [1] "name"                 "state.x"              "state_code"
##  [4] "type"                 "degree_length"        "room_and_board"
##  [7] "in_state_tuition"     "in_state_total"       "out_of_state_tuition"
## [10] "out_of_state_total"   "system"               "total_enrollment"
## [13] "state.y"              "category"             "enrollment"
## [16] "share"
```

Now we'll actually do the join.

```r
comb_df = inner_join(div_df, tuition_df, by = c('name', 'state'))
```

1.5 School Diversity Violin Plots

Because of the way `div_df` is structured, it makes sense to separate all of the plots for it into separate panels using the `category` variable. We've used `facet_wrap()` in this way before.

```
ggplot(comb_df, aes(system, share, color = system)) +
    geom_violin(scale = 'width',
                draw_quantiles = .5) +
    geom_beeswarm(alpha = .25, cex = 2) +
    facet_wrap(vars(category), scales = 'free')
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
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## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```
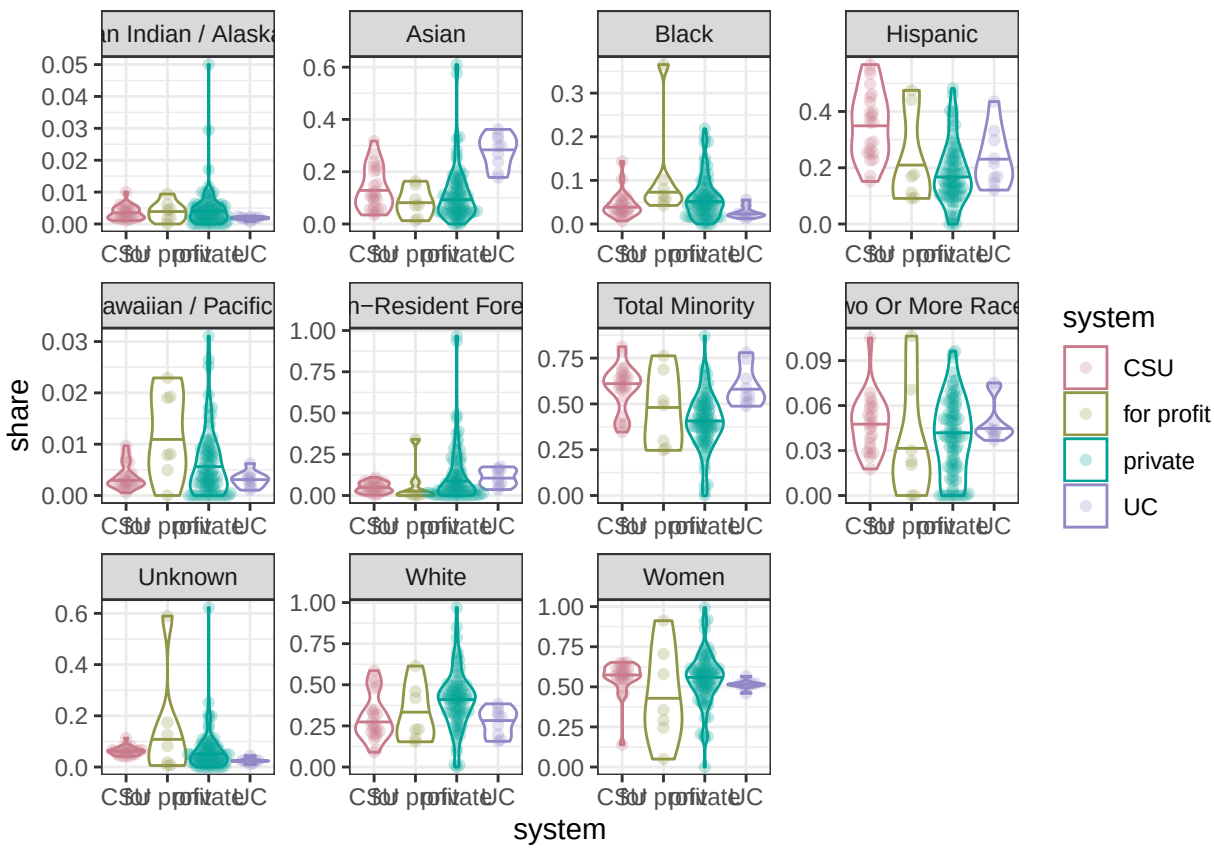



- No school in the data has more than 5% American Indian/Alaska Native students.

- The numbers are even lower for Native Hawaiian/Pacific Islanders.
- Just a few schools have more than 10% Black students.

- The median UC and CSU schools have greater "Total Minority" students than the private schools.
  - But there's a lot of variation within the CSU system and across private schools.
- UC and CSU schools have different groups of minority students

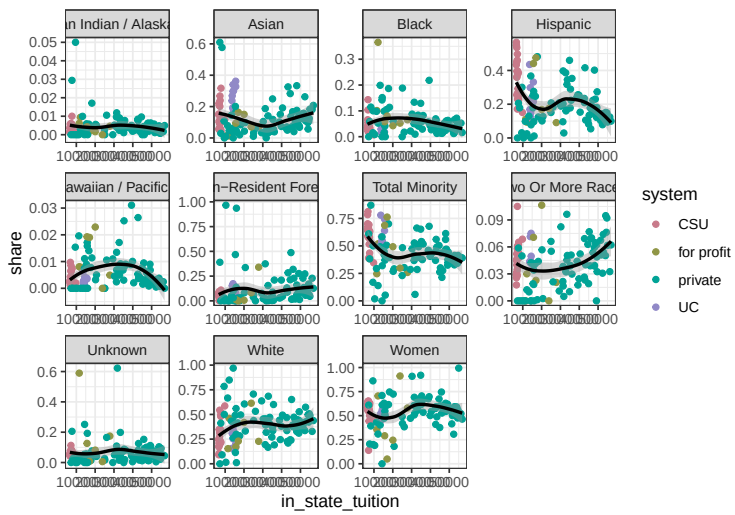
- UCs have a higher share of Asian students than Cal States
- Cal States have a higher share of Hispanic students than UCs

1.6 2D Distributions: Numeric x Numeric

A point geom gives us a basic scatterplot. `geom_smooth()` adds a predictive model. By default, this predictive model is either a LOESS (also called local regression; for smaller datasets) or a generalized additive model (GAM; for larger datasets).

```
ggplot(comb_df, aes(in_state_tuition, share)) +
    geom_point(aes(color = system)) +
    geom_smooth(color = 'black') +
    facet_wrap(vars(category), scales = 'free')
```

```
## `geom_smooth()` using method = 'loess' and formula 'y ~ x'
```



1.6.1 Discussion question

Incorporating a regression model into an exploratory analysis like this will almost always be a fraught decision. On the one hand, it aids pattern discovery, and isn't necessarily a radical step beyond plotting the data if we already think of the plot itself as a model. On the other hand, any regression model involves substantive assumptions, and it can be difficult to interrogate those assumptions when the model is used in this way.

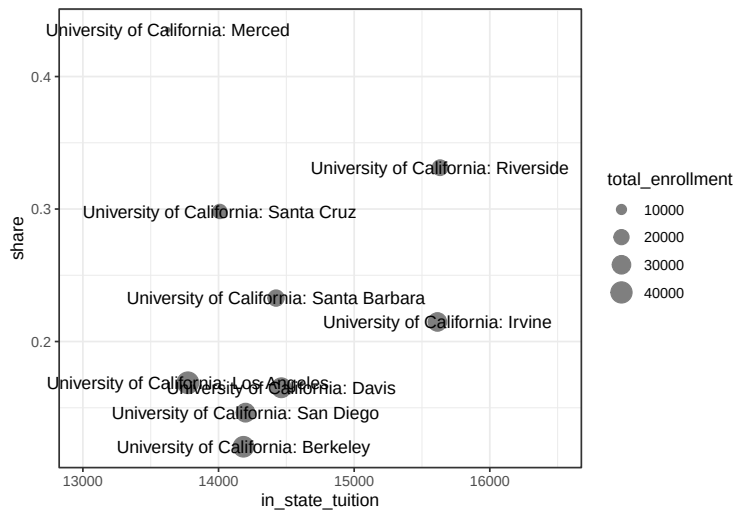
What other considerations should we take into account when deciding whether to add a model to an exploratory plot?

1.7 Text geoms

The final plot highlights how easy it is to extend a traditional scatterplot using ggplot. First, the `text` geom prints strings of text (see also the `label` geom). This is especially useful when we want to identify individual observations but have too many values for colors or shapes. To accommodate the long labels, we expand the range of the x-values in the coordinate system and turn off clipping at the plot margins. We also scale the size of the points to campus size, which suggests a correlation between size (which in this case is tightly correlated with age and prestige) and Hispanic share. Aesthetics can be handled for the entire plot or separately for each layer, as needed to create effective cognitive affordances.

```
comb_df %>%
    filter(system == 'UC', category == 'Hispanic') %>%
    ggplot(aes(in_state_tuition, share)) +
```

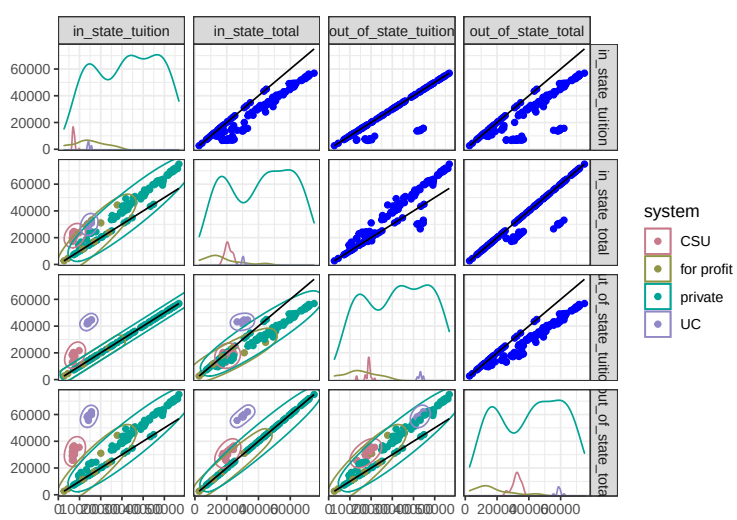
```
    geom_point(aes(size = total_enrollment), alpha = .5) +
    geom_text(aes(label = name)) +
    coord_cartesian(clip = 'off', xlim = c(13000, 16500))
```



1.8 Multivariate distributions

- (This should probably get moved to the next chapter, but I'll leave it here just in case we make it this far in class.)

```
ggplot(tuition_df, aes(.panel_x, .panel_y, color = system)) +
    geom_autopoint(color = 'blue') +
    geom_autodensity(fill = NA, position = 'identity') +
    geom_autopoint() +
    geom_mark_ellipse(expand = .05) +
    stat_function(fun = identity, inherit.aes = FALSE) +
    facet_matrix(vars(in_state_tuition, in_state_total,
                      out_of_state_tuition, out_of_state_total),
                 layer.upper = c(1, 5),
                 layer.diag = 2,
                 layer.lower = c(3, 4, 5))
```



1.9 Lab

Once you're familiar with joins and the basics of ggplot, you're ready to tackle the next lab, which is available here: https://github.com/data-science-methods/lab_w06_covid